USN | | | | | | | | | | | [22TCSE103]

FIFTH SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

COMPUTER NETWORKS

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	List out any five features of layered architecture.	L2	CO1	**02**
	b.	Outline the responsibilities of a Transport Layer.	L2	CO2	**02**
	c.	Summarize the limitations of current IPV4 protocol.	L2	CO3	**02**
	d.	Differentiate between Pure Aloha and Slotted Aloha.	L2	CO4	**02**
	e.	How does frame Relay is working in packet-switching network protocol?	L2	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Classify the application Layer Protocol. Explain with an example.	L2	CO1	**05**
		OR			
	b.	Describe the function of the Domain Name System(DNS) within the TCP/IP suite.	L2	CO1	**05**
3.	**a.**	How does TCP ensure data reliability during transmission? Explain in detail.	L4	CO2	**05**
		OR			
	b.	Illustrate the TCP Connection Establishment mechanism with diagram.	L2	CO2	**05**
4.	**a.**	Explain the differences between static and dynamic IP addressing. List out the advantages and disadvantages of each method.	L4	CO3	**05**
		OR			
	b.	Explain how DHCP protocol assigned the IP address dynamically in network management.	L3	CO3	**05**
5.	**a.**	Explain how to transfer the data between two devices using CSMA method in communication channel.	L4	C04	**05**
		OR			
	b.	Explain the role of NAT (Network Address Translation) in mitigating IPv4 limitations. What are the drawbacks of relying on NAT?	L2	CO4	**05**
6.	**a.**	Illustrate the Inter-VLAN Routing process with a neat diagram.	L2	CO5	**05**
		OR			
	b.	How wireless Router Services is converged in WAN network? Explain in detail.	L4	CO5	**05**

7.	**a.**	Discuss the role of IPv6 in supporting Internet. What are the features of IPv6 is suitable for Internet applications?	L2	CO6	**05**
		OR			
	b.	How security provided for 802.11 networks? Explain in detail.	L2	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Illustrate the functions of HTTP, FTP, SMTP protocols.	L2	CO1	**10**
		OR			
	b.	Explain the types of Domain Name System with suitable examples.	L2	CO1	**10**
9.	**a.**	Explain how TCP congestion control is a method used by the TCP protocol to manage data flow over a network and prevent congestion.	L4	CO2	**10**
		OR			
	b.	Illustrate how UDP enables faster transmission as a connectionless protocol.	L4	CO2	**10**
10.	**a.**	Explain the working principles of Distance Vector routing algorithm with neat diagram.	L2	CO3	**10**
		OR			
	b.	Discuss about Packet Forwarding and Routing process in IPv4 Networks.	L4	CO3	**10**
11.	**a.**	Explain the requirements for VLAN Trunking Protocol to communicate the information between switches.	L2	CO5	**10**
		OR			
	b.	Explain about Point-to-Point Protocol frame format with various fields.	L4	CO5	**10**
12.	**a.**	Explain how VLSM technique used in IP network design to create subnets with different subnet masks.	L2	CO3	**10**
		OR			
	b.	How RIPv1 protocol used as an intra domain routing which is based on distance vector routing inside an autonomous system. Explain in detail.	L4	CO3	**10**
13.	**a.**	How Spanning Tree Protocol is used to make a loop free network? Explain in detail.	L4	CO4	**10**
		OR			
	b.	Explain how Real Time Transport Protocol is designed to handle real-time traffic of the Internet.	L4	CO6	**10**